

Michael Rubin MEng Mechanical Engineer

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Profile Summary

Driven MechE graduate with hands-on experience. Motivated to apply my technical depth and creative problem-solving in a **design or manufacturing role**. With projects taken from analysis to functional hardware under tight timelines, I bring a strong background in **engineering analysis, rapid prototyping, design optimization and troubleshooting of electromechanical systems**. Fluent in **5 languages**, I collaborate effectively across multidisciplinary teams in fast-paced environments.

Experience

Mechanical/Controls Engineer - Project LUCI | MPC Lab, UC Berkeley – Berkeley, CA, USA May – Present

- Contract role supporting NIWC Pacific on development of an **all-terrain autonomous surveillance rover**.
- Redesigning 3D-printed chassis** of the rover to improve structural robustness, manufacturability, and usability.
- Designing and executing **testing protocols** to validate control architecture; integrating NTRIP GPS + vSLAM navigation.

Chassis Engineer - Driver Safety Lead | CALSOL Solar Vehicle Team – Berkeley, CA, USA Sept 2025 – Present

- Working on GenXI seatbelt implementation in the occupant cell, ensuring ASC and WSC **regulation compliance**.
- Lap and sub belt mounts **weight reduced by 15%**. Designed test proving mounts withstand crashes load with 1.6 FOS.
- Lowered production complexity** of the shoulder belt anchorage enabling in-house manufacturing.

AutoCAD Drawing Engineer | KBC Acurity – Remote Oct 2022 – June 2025

- Produced over 300 **technical drawings** for nationwide security camera installations at KBC Bank offices.

Teaching Assistant (UGSI: Statics, Mechanics of Materials) | TU Delft – Delft, ZH, The Netherlands Sept 2022 – Jan 2023

Projects

Handheld Autorefracton Device | UC Berkeley - Continued Independently Jan 2025 – Present

- Led optical design** of a Scheiner disk system; designed to $\pm 0.50^\circ$ SPH accuracy. Secured \$1K grant for prototyping.
- Created alignment jigs, calibration protocols and an image-processing pipeline to compute corrective lens power.

Robotic End Effector, Graduate Capstone | Embodied Dexterity Group – UC Berkeley Sept 2025 – May 2026

- Improved manufacturability** of a multi-chamber suction cup end-effector that enables haptic feedback.
- Scaled the design from a research-grade prototype to a **production-ready unit for 1,000+ orders**.
- Standardized robot-arm interface; cut part count 32→9, setup time 15min→1:45min; designed & integrated custom PCB.

Light module for Robotic Surgery System, BSc Thesis | AdLap Lab – TU Delft Jan 2024 – June 2024

- Designed **OR compliant** detachable light module for a low-cost laparoscopic system, with white/IR mode switching.
- Increased illuminance 3x; halved device size; **handled thermal management**; validated via thermal & optical testing.

CNC machine | TU Delft Sept 2023 – Dec 2023

- Designed and built a CNC machine using an H-bot single-belt configuration; Placed top 5% in faculty competition.
- Cut part count by 11 and footprint by 10% vs. 3-axis systems; Integrated belt pre-tensioning and vibration damping.

Relevant Skills

Mechanical: CAD & Drawing (SolidWorks CSWA certified, OnShape), FMEA, GD&T (ASME Y14.5-2018), DFx, RCA, V&V, Material Selection (Granta), BOM Management, Motor & Sensor Selection, Thermal optimization, Control Systems

Manufacturing: CNC, Additive (FFF, DLP, Cura), Injection Molding, Sheet Metal Fab, Bench Tests, Machine Shop Trained

Software: FEA (Abaqus, Ansys, COMSOL), Programming (Python, MATLAB, C++, G-Code), Git, Ros2, VSCode, Linux

Languages & Soft Skills: English (near-native, TOEFL 114/120), French (native), Dutch (native), Spanish (fluent), Hebrew (elementary), Adaptability, Reporting & Documentation, Collaboration, Task Prioritization, Creativity

Education

University of California at Berkeley (UC Berkeley) – GPA 3.8/4 Berkeley, CA, USA

Master's degree in Mechanical Engineering – Biomechanics & Control Aug 2025 – May 2026

Delft University of Technology (TU Delft) – GPA 3.8/4 Delft, The Netherlands

Honours Bachelor's degree in Mechanical Engineering Aug 2021 – June 2024

- Exchange semester at Norwegian University of Science and Technology (NTNU) – GPA 3.825/4

Awards & Honors

Graduate Scholarship | Belgian American Education Foundation Aug 2025 – May 2026

One of 96 Belgians awarded a national grant to pursue graduate studies in the United States; covering full tuition & stipend.

CERN Honours Summerschool Travel Grant | CERN IdeaSquare May 2024 – Aug 2024

Awarded competitive travel grant covering costs and accommodation of the program; granted to top 25% of cohort.

Honors Program Delft Certificate | TU Delft Sept 2022 – June 2024

Selected as 1 of 4 MechE students for exclusive program; completed 20 additional credits within nominal graduation time.